

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: A Personalized News Feed Recommender (e.g., Twitter/X or TikTok Algorithm)

PEAS Specification

A Personalized News Feed Recommender uses machine learning to rank and recommend posts, videos, and articles that best match an individual user's interests and engagement patterns. The system continuously adapts its recommendations based on user interactions and contextual information.

Performance Measures

- Click-Through Rate (CTR) on recommended content.
- User Engagement Time (watch time / reading time).
- Recommendation Relevance Accuracy based on user feedback.
- User Retention and Daily Active User (DAU) growth.
- Diversity of recommended content to avoid filter bubbles.
- Low rate of irrelevant or hidden/reported recommendations.
- Average recommendation response latency.

Environment

The recommender operates on social media and content-sharing platforms across mobile apps and websites. It interacts with millions of users, creators, advertisers, and content databases in a highly dynamic environment where new content is uploaded continuously and user interests change over time.

Actuators

- Feed Ranking Engine that orders posts and videos.
- Content Recommendation Module that suggests new creators and topics.
- Notification System that pushes personalized alerts.
- Advertisement Selection Engine for targeted promotions.
- Content Filtering Module that suppresses harmful or duplicate content.

Sensors

- User interaction logs (likes, shares, comments, clicks).
- Watch time and scrolling behavior trackers.
- User profile information and follow history.
- Device, location, and time-of-day context signals.
- Content metadata (hashtags, categories, keywords).
- Feedback signals such as 'Not Interested' or reports.

Environment Classification

Dimension	Classification	Justification
Observability	Partially Observable	The system cannot directly know a user's true preferences and only infers them from behavior.
Agency	Multi-agent	It interacts with millions of users, creators, advertisers, and other recommendation systems.
Determinism	Stochastic	User behavior and content popularity are unpredictable.
Episode Structure	Sequential	Current recommendations depend on previous interactions and history.
Temporal Dynamics	Dynamic	New content and user interests change continuously.
State Space	Continuous	User preferences and engagement signals evolve over a large continuous feature space.

Critical Analysis

Greatest Technical Challenge

The partially observable nature of the environment is the greatest challenge. A recommender system never has direct access to a user's actual interests or intentions. It only observes indirect signals such as clicks, watch time, or likes, which may not always reflect genuine preference. The system must therefore estimate hidden user preferences while adapting to changing trends and avoiding overfitting to short-term behavior.

Utility Function and Trade-off

$$U = 0.5 \times \text{Engagement_Score} + 0.3 \times \text{Relevance_Score} + 0.2 \times \text{Diversity_Score}$$

This utility function prioritizes maximizing engagement while also rewarding accurate recommendations and maintaining content diversity. If the engagement weight were doubled, the recommender would favor highly addictive or viral content, increasing user activity but potentially reducing diversity and exposing users to repetitive recommendations.

Structured Representation (JSON)

```
{
  "system_name": "Personalized News Feed Recommender",
  "peas": {
    "performance_measures": ["Click-Through Rate", "User Engagement Time", "Recommendation Accuracy", "User Retention", "Content Diversity"],
    "environment": "Dynamic social media platform with continuously changing users and content",
    "actuators": ["Feed ranking engine", "Recommendation module", "Notification system", "Ad selection engine"],
    "sensors": ["User interactions", "Watch time", "Profile data", "Content metadata", "Feedback signals"]
  },
  "environment_classification": {
    "observability": {"choice": "Partially Observable", "justification": "True user preferences cannot be directly observed."},
    "agents": {"choice": "Multi-agent", "justification": "Many users and content providers interact simultaneously."},
    "determinism": {"choice": "Stochastic", "justification": "User responses are inherently unpredictable."},
    "episode_structure": {"choice": "Sequential", "justification": "Past interactions influence future recommendations."},
    "temporal_dynamics": {"choice": "Dynamic", "justification": "Content and preferences change constantly."},
    "state_space": {"choice": "Continuous", "justification": "Preference and engagement values vary continuously."}
  },
  "utility_function": "U = 0.5*Engagement + 0.3*Relevance + 0.2*Diversity"
}
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